

CS & IT ENGINEERING



Computer Network - 2

MAC Sub-Layer

Lecture No. - 05

By - Abhishek Sir





Recap of Previous Lecture



Topic

CSMA ✓

Topic

CSMA/CD ✓

Topic

Ethernet



Topics to be Covered



Topic

Ethernet

Topic

Back-off Algorithm

Topic

Ethernet Bridging

[MCQ]

[GATE-2003] [2 Mark]



#Q. A 2 km long broadcast LAN has 10^7 bps bandwidth and uses CSMA/CD. The signal travels along the wire at 2×10^8 m/s. What is the minimum packet size that can be used on this network ?

- ☐ A 50 bytes
- ☐ B 100 bytes
- ☐ C 200 bytes
- ☐ D None of the above

Solution :

$$\begin{aligned}
 \text{Minimum frame size} &= 2 * (\text{Distance} / \text{Signal Speed}) * \text{Bandwidth} \\
 &= 2 * (2 \text{ km} / 2 * 10^8 \text{ m/sec}) * 10^7 \text{ bits/sec} \\
 &= 2 * (2 * 10^3 \text{ m} / 2 * 10^8 \text{ m/sec}) * 10^7 \text{ bits/sec} \\
 &= 200 \text{ bits} \\
 &= 25 \text{ bytes}
 \end{aligned}$$

[MCQ]

[GATE-IT-2005] [2 Mark]



#Q. A network with CSMA/CD protocol in the MAC layer is running at 1 Gbps over a 1 km cable with no repeaters. The signal speed in the cable is 2×10^8 m/sec. The minimum frame size for this network should be

- ☐ A 5000 bits
- ☐ B 5000 bytes
- ☐ C 10000 bits
- ☐ D 10000 bytes

Solution :

$$\begin{aligned}
 \text{Minimum frame size} &= 2 * (\text{Distance} / \text{Signal Speed}) * \text{Bandwidth} \\
 &= 2 * (1 \text{ km} / 2 \times 10^8 \text{ m/sec}) * 1 \text{ Gbps} \\
 &= 2 * (10^3 \text{ m} / 2 \times 10^8 \text{ m/sec}) * 10^9 \text{ bits/sec} \\
 &= 10000 \text{ bits} \\
 &= 1250 \text{ bytes}
 \end{aligned}$$

[MCQ]

[GATE-IT-2008] [2 Mark]



#Q. The minimum frame size required for a CSMA/CD based computer network running at 1 Gbps on a 200 m cable with a link speed of 2×10^8 m/s is :

- ☐ A 125 bytes
- ☐ B 250 bytes
- ☐ C 500 bytes
- ☐ D None of these

Solution :

$$\begin{aligned}
 \text{Minimum frame size} &= 2 * (\text{Distance} / \text{Signal Speed}) * \text{Bandwidth} \\
 &= 2 * (200 \text{ m} / 2 \times 10^8 \text{ m/sec}) * 1 \text{ Gbps} \\
 &= 2 * (200 \text{ m} / 2 \times 10^8 \text{ m/sec}) * 10^9 \text{ bits/sec} \\
 &= 2000 \text{ bits} \\
 &= 250 \text{ bytes}
 \end{aligned}$$

[NAT]

[GATE-2016] [2 Mark]



#Q. A network has a data transmission bandwidth of 20×10^6 bits per second. It uses CSMA/CD in the MAC layer. The maximum signal propagation time from one node to another node is 40 microseconds. The minimum size of a frame in the network is _____ bytes.

Solution :

$$\begin{aligned}
 \text{Minimum frame size} &= 2 * t_p * \text{Bandwidth} \\
 &= 2 * 40 \text{ microsecond} * 20 * 10^6 \text{ bits / second} \\
 &= 2 * 40 * 10^{-6} \text{ second} * 20 * 10^6 \text{ bits / second} \\
 &= 1600 \text{ bits} \\
 &= 200 \text{ bytes}
 \end{aligned}$$

[MCQ]

[GATE-2015] [1 Mark]



#Q. Consider a CSMA/CD network that transmits data at a rate of 100 Mbps (10^8 bits per second) over a 1 km (kilometer) cable with no repeaters. If the minimum frame size required for this network is 1250 bytes, what is the signal speed (km/sec) in the cable ?

- A 8000
- B 10000
- C 16000
- D 20000

Solution :

$$\begin{aligned}
 \text{Minimum Signal Speed} &= 2 * \text{Distance} * (\text{Bandwidth} / \text{Frame size}) \\
 &= 2 * 1 \text{ km} * (100 \text{ Mbps} / 1250 \text{ bytes}) \\
 &= 2 * 1 \text{ km} * (10^8 \text{ bits per second} / 10^4 \text{ bits}) \\
 &= 20000 \text{ km per second}
 \end{aligned}$$

[NAT]

[GATE-2024] [2 Mark]



#Q. Consider an Ethernet segment with a transmission speed of 10^8 bits/sec and a maximum segment length of 500 meters. If the speed of propagation of the signal in the medium is 2×10^8 meters/sec, then the minimum frame size (in bits) required for collision detection is _____ .

Solution :

$$\begin{aligned}
 \text{Minimum frame size} &= 2 * (\text{Distance} / \text{Signal Speed}) * \text{Bandwidth} \\
 &= 2 * (500 \text{ meters} / 2 \times 10^8 \text{ meters/sec}) * 10^8 \text{ bits/sec} \\
 &= 500 \text{ bits}
 \end{aligned}$$

[MCQ]

[GATE-2013] [2 Mark]



#Q. Determine the maximum length of the cable (in km) for transmitting data at a rate of 500 Mbps in an Ethernet LAN with frames of size 10,000 bits. Assume the signal speed in the cable to be 2,00,000 km/s.

- ☐ A 1
- ☐ B 2
- ☐ C 2.5
- ☐ D 5

Solution :

$$\begin{aligned}
 \text{Maximum Distance} &= (\text{Frame size} / \text{Bandwidth}) * (\text{Signal Speed} / 2) \\
 &= (10000 \text{ bits} / 500 \text{ Mbps}) * (200000 \text{ km/sec}) / 2 \\
 &= (10^4 \text{ bits} / 5 * 10^8 \text{ bits/sec}) * (2 * 10^5 \text{ km/sec}) / 2 \\
 &= 2 \text{ km}
 \end{aligned}$$

#Q. In a computer network using CSMA/CD, the minimum frame size for data transfer is 1500 bytes. What will be the data transfer speed in this network over 1 Km cable, if the signal speed in the cable is 250000 Km/sec and no repeaters are used in this network ?

- A 100 Mbps
- B 500 Mbps
- C 1 Gbps
- D 1.5 Gbps



Topic : Jam Signal

⇒ continuous one's



→ Jamming signal [Minimum size 32-bits or 48-bits]

→ Transmitter start transmitting jam signal when collision detected

→ To inform the other transmitting stations about the collision

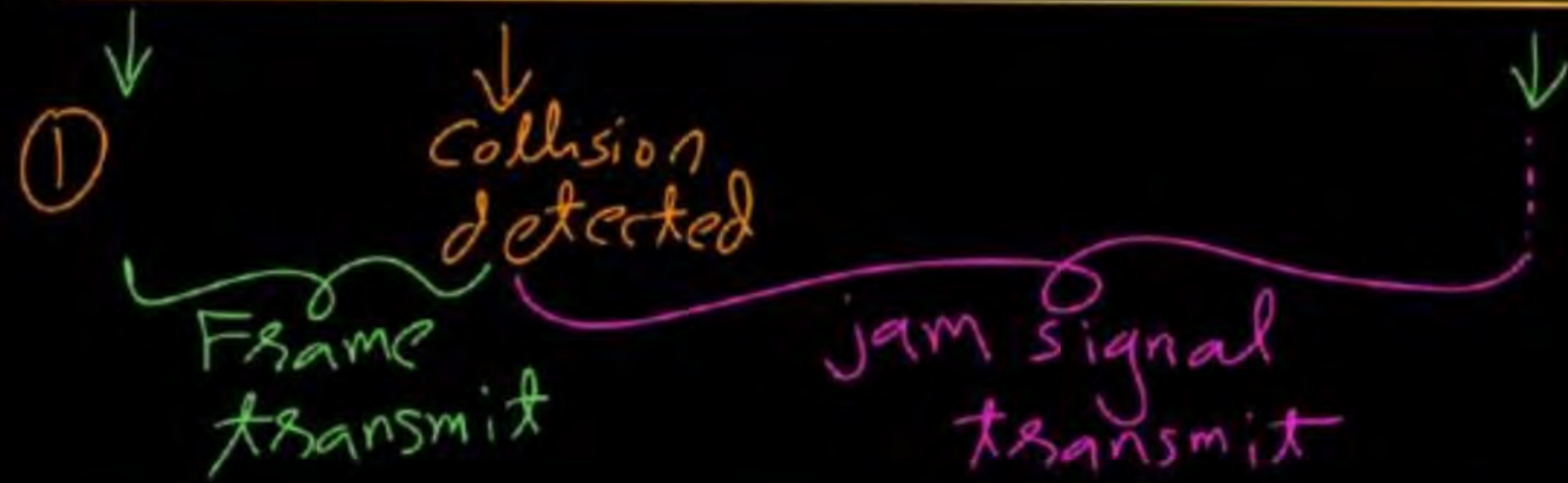
$$\underline{t_x} \geq [2 * t_p + t_{xj}]$$

$$\begin{aligned} \text{Min}^m \text{ transmission} \\ = (2t_p + t_{xj}) \end{aligned}$$

t_{xj} : Transmission time for Jamming Signal (for min^m size jam signal)

[If no any information given about jam signal, consider negligible size]

0 $(2t_p)$ $(2t_p + t_{xj})$ t_x



[MCQ]

[GATE-CS-2005][2 Mark]



#Q. Suppose the round trip propagation delay for a 10 Mbps Ethernet having 48-bit jamming signal is 46.4 microseconds. The minimum frame size is :

- ☐ A 94 bits $2t_p = 46.4 \mu s$
- ☐ B 416 bits
- ☐ C 464 bits $\Rightarrow [t_x \geq 2t_p]$
- ☒ D 512 bits $\Rightarrow t_x \geq [2t_p + t_{xj}]$

Ans: D

Solution :

$$t_x = [2 * t_p + t_{xj}]$$

Minimum frame size = $[2 * t_p + t_{xj}] * \text{Bandwidth}$

$$= \left[46.4 \mu s + \frac{48 \text{ bit}}{10 \text{ Mbps}} \right] * 10 \text{ Mbps}$$

$$= 46.4 \mu s * 10 \text{ Mbps} + 48 \text{ bit}$$

$$= (46.4 * 10^{-6} \text{ sec} * 10 * 10^6 \text{ bits/sec}) + 48 \text{ bits}$$

$$= (464 \text{ bits} + 48 \text{ bits})$$

$$\text{Min}^m \text{ Frame Size} = 512 \text{ bits} = 64 \text{ bytes}$$

Solution :

$$t_x = [2 * t_p + t_{xj}]$$

$$\begin{aligned} \text{Minimum frame size} &= [2 * t_p + t_{xj}] * \text{Bandwidth} \\ &= [46.4 \mu\text{s} + (\text{Jam Signal Size} / \text{Bandwidth})] * \text{Bandwidth} \\ &= [46.4 \mu\text{s} * \text{Bandwidth} + \text{Jam Signal Size}] \\ &= [46.4 \mu\text{s} * 10\text{Mbps} + 48 \text{ bits}] \\ &= [46.4 \mu\text{s} * 10^7 \text{ bits/sec} + 48 \text{ bits}] \\ &= [464 \text{ bits} + 48 \text{ bits}] \\ &= 512 \text{ bits} \\ &= 64 \text{ bytes} \end{aligned}$$

#Q. In Ethernet CSMA/CD, the special bit sequence transmitted by media access management to handle collision is called

- ☒ A Preamble
- ☒ B Postamble
- ☒ C Jam
- ☐ D None of the above

Ans: C



Topic : Exponential Back-off Algorithm

-> Binary Exponential Back-off Algorithm

-> At k^{th} collision of particular frame : $[k = 1, 2, 3, \dots]$

if $k < 15$

then transmitter chooses a number R randomly
in between 0 to $(2^i - 1)$ where $i = \min(k, 10)$

else

Abort the re-transmission





Topic : Exponential Back-off Algorithm

-> Wait Time = ~~$R * t_x$~~ $R * \text{Min}^m \text{ transmission time}$

-> Transmitter will sense the channel after Wait Time
[for re-transmission of the frame]

$$\text{Min}^m \text{ transmission time} = (2t_p + t_x)$$



Topic : Exponential Backoff Algorithm



k^{th} collision	Range
1 st	0, 1
2 nd	0, 1, 2, 3
3 rd	0 - - - - - 7
4 th	0 - - - - - 15
⋮	
9 th	0 - - - - - 511
10 th	0 - - - - - 1023
11 th	0 - - - - - 1023
15 th	Abort

Example 6 :-

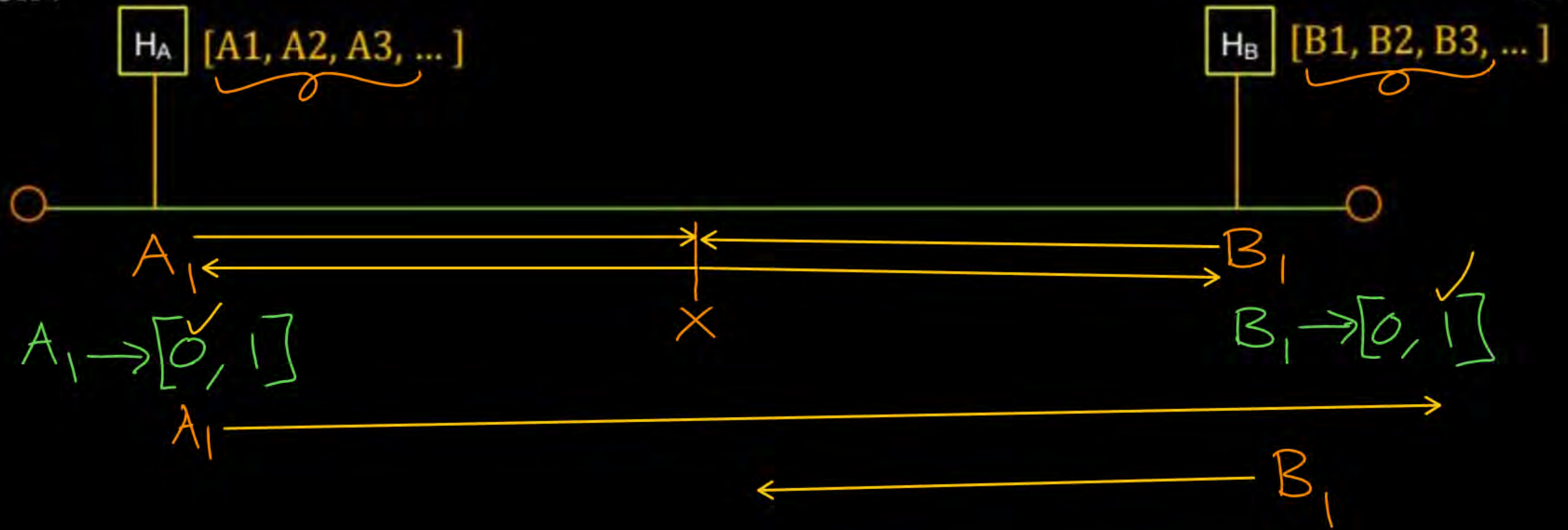
[NAT]



#Q. Consider A and B are the only two stations on an Ethernet LAN. Each has a steady queue of frames to send. Both A and B attempt to transmit a frame and collide, what is the probability that A wins the first back-off race?



Solution :-



A1	B1	Result
0	0	} Again Collision
1	1	
0	1	A win
1	0	B win

$$\text{Ans} = \frac{1}{4} = 0.25$$

[MCQ]

11T-7, H.W.

[GATE-CS-2004][2 Mark]



#Q. A and B are the only two stations on an Ethernet. Each has a steady queue of frames to send. Both A and B attempt to transmit a frame, collide, and A wins the first back-off race. At the end of this successful transmission by A, both A and B attempt to transmit and collide. The probability that A wins the second back-off race is :

- ☐ A 0.5
- ☐ B 0.625
- ☐ C 0.75
- ☐ D 1.0

[MCQ]

CSMA/CD

IISC,

[GATE-2016] [1 Mark]



#Q. In an Ethernet local area network, which one of the following statements is TRUE?

- ☒ A A station stops to sense the channel once it starts transmitting a frame. T-CSMA
F-CSMA/CD
- ☒ B The purpose of the jamming signal is to pad the frames that are smaller than the minimum frame size. FALSE
- ☒ C A station continues to transmit the packet even after the collision is detected. F
- ☒ D The exponential back-off mechanism reduces the probability of collision on re-transmissions. TRUE

Ans: D



2 mins Summary



Topic

Ethernet

Topic

Back-off Algorithm

Topic

~~Ethernet Bridging~~



THANK - YOU

